

Workshop Python Image Analysis

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Chapter I: About, goals, setting up & introduction

About

There exist many good image analysis tutorials. I wrote this one because it's easier to teach my own texts, and it aims to follow the *Carpentries* style code-along.

In this workshop, there are chunks of code that are intended to be typed by the instructor, to be projected on a large screen. Workshop participants should type along, in order to get “hands on” experience.

In addition, there are exercise parts, which participants can do on their own, supervised by the instructor and helpers. Some parts are notebooks on which you can work completely on your own.

Goals and philosophy

The goals of this python-based workshop are to teach you:

- How to think about images on the computer
- Learn about tools to modify images
- Which tools are available in Python
- Python syntax/concepts required to apply these tools
- How to practically build an image analysis-pipeline
- Some fundamentals of ML

The idea is *not* to train you in every single method that exists in image analysis. We go over main ideas of image analysis, and the basics of how to manipulate images.

Once you know what types of possibilities and tools to look for, you can further explore them on your own using internet resources. Querying Google, or asking “Claude”, “Chat” or “Gem” (ie Large Language Models, or LLMs) for advice is very useful when programming. This course is designed with the idea in mind that many further resources are available to help you build the specific pipeline you need. This workshop intends to give you an overview of what’s out there and a solid foundation to build on.

The bioDSC

When you have your own particular challenges and want more specific advice, don’t hesitate to contact me or my colleagues at the biological Data Science Center for advice at any time.

See <https://www.biodsc.nl/>.

Requirements

Regarding your experience:

- We require you to have followed our “introduction to python” course, OR ..
- .. require you to be familiar with the topics we taught there.

On your computer, we require you to have **before the workshop starts**:

- The software “FIJI” ([link](#)).
- Conda software:
 - See this [link](#) (section “How to install Conda”).
 - * You can stop once you’ve installed ‘conda’.
 - * The rest of the linked website provides some background.
- A ‘conda environment’ with the libraries listed below installed (see section below, “[Libraries to install](#)”).
- An editor in which you can edit and run python code
 - For this advanced workshop we require an IDE (integrated development environment)
 - We’ll be using “Spyder” during the workshop.
 - To install Spyder and your conda environment:
 - * [follow instructions here](#) (section “standalone installers”).

- * Now find out where Conda installed Python. Start the terminal/powershell/prompt. Activate your conda environment by `conda activate 2026_py_image_workshop`. Then,
 - **On a macbook:** type `which python`, to find out the path.
 - **On Windows:** start Python using the `python` command. Now type the commands `import sys` and `print(sys.executable)` to find out the path.
- * In Spyder, open preferences (pythonX.XX > preferences OR Tools > Preferences), go to ‘Python interpreter’, select ‘Selected interpreter’, and choose the python path identified in the previous step. (See also [this link](#).)
- If you’re familiar with Visual Studio Code or Pycharm, you can use those IDEs as well.
- Download the datasets linked below (see section below, “[Datasets to download](#)”).

If you have trouble with installing any of these, **please contact us before the workshop!**

Libraries to install

You can create and activate an environment, plus install the required libraries, with the commands below.

```
# Create and activate conda environment
conda create -n 2026_py_image_workshop
conda activate 2026_py_image_workshop
# Add "conda-forge" to default channels
conda config --add channels conda-forge
# Libraries to install
conda install matplotlib seaborn tifffile numpy scikit-image scipy imageio napari pandas pyyaml

# Optional: ipykernel (useful when using VS Code)
conda install ipykernel

# You can choose to also install pytorch (doesn't work on all computers)
# or use google colab for that chapter.
conda install pytorch -c pytorch
```

To use this environment from the terminal, open a new terminal window, and type `conda activate 2026_py_image_workshop`.

See also [this blog post](#) to find out how to install Conda.

Known issues and solutions

Conda needs updating

Sometimes, conda needs to be updated before commands work. Use:

```
conda update conda
```

Installing all packages at once not working

Sometimes Napari will give installation issues. Or sometimes other packages might give issues. Try installing one by one.

Napari might need to be installed using the following code (use these commands after activating the environment via `conda activate 2026_py_image_workshop`):

```
python -m pip install "PyQt5<5.16"
python -m pip install "napari[pyqt5]"
```

This will explicitly install the PyQt5 dependency and then Napari.

Things going very slowly

If installation is extremely slow, you can try to use a much faster version of Conda, which is called “Mamba”.

Install it first (after activating your environment via `conda activate 2026_py_image_workshop`):

```
conda install mamba
```

After this, you can replace any “conda” in your command with “mamba”, e.g.

```
# Conda command:
conda install pandas
# Equivalent but faster:
mamba install pandas
```

Google Colab

There is a Google Colab document, which allows you to perform this workshop online:

See: https://colab.research.google.com/drive/1qJgibG2-ACfXJZky_VMP8Wjm3quhpgY_?usp=sharing

I recommend using your own IDE, such that you can later easily analyze your own data. However, for the section on machine learning, this might be necessary, as not all platforms easily allow pytorch installation and configuration.

For reference: specifying the (Colab) environment

You can set up a new colab document using the following code:

```
# These are likely already installed on Google Colab:
!pip install numpy matplotlib seaborn scipy torch pandas

# These might need to be installed explicitly:
!pip install scikit-image tifffile imageio napari pyyaml
```

Datasets to download

- Example images;
 - <https://surfdrive.surf.nl/s/m2GE3GBwaxWKSDF>
- KTR example data;
 - <https://surfdrive.surf.nl/s/dHCHgk66DTd7Hnz>
- Bacterial filamentation data
 - <https://surfdrive.surf.nl/s/FHaHjGWRNTxxzLT>

Regarding the use of LLMs

Given the omnipresence of LLMs/AIs, a word of advice related to using them.

Large Language Models (LLMs) can be a very useful tool in building your image analysis pipeline. I think they are most useful when you already have a good level of knowledge of a specific topic (bio-image analysis and python scripting in this case). In that case, you know which questions to ask, and can assess the answers you get for correctness. When you ask them to generate code, you can assess whether the code makes sense, or is bloated with unnecessary parts, or is just plain wrong – the latter two are a common occurrence!.

If you want to become (intermediately) proficient in (bio-image analysis) coding, I strongly advise you to *not* use AI to write all your code. To gain skill, at least spend 50% of your time spend on scripts writing your own code (you can ask AI questions ofcourse).

During this course, it's strongly advised to not let AI generate code for you.

Contact us

In case of any problems, please do contact us (info@biodsc.nl)!

Acknowledgements

This course is (very loosely) based on <https://bioimagebook.github.io/>, which is available under the [Creative Commons license](#).

We'll be using two example image sets for this tutorial, plus some additional single examples. Those two sets come from:

- Sergei Chavez-Abiega et al. (2022), “Single-cell imaging of ERK and Akt activation dynamics and heterogeneity induced by G-protein-coupled receptors”, Journal of Cell Science, <https://doi.org/10.1242/jcs.259685>
- Martijn Wehrens et al. (2018), “Size Laws and Division Ring Dynamics in Filamentous Escherichia Coli Cells”, Current Biology, <https://doi.org/10.1016/j.cub.2018.02.006>.

Let's start: Introduction

Organization

- Slides;
 - Introducing the bioDSC
 - Introduce yourselves
 - Introducing Carpentries style teaching.

Why use python to analyze images?

(Ask participants what they think.)

- Biological data can be generated at scale
- Automated analysis
 - Save human labor, scalable
 - Transparent, reproducible
 - Unbiased
 - Highly customizable